

Wireshark Tasks

1. The Basic HTTP GET/response interaction:

- Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?

```
Request Version: HTTP/1.1
```

HTTP 1.1

- What languages (if any) does your browser indicate that it can accept to the server?

```
Accept-Language: en-GB,en;q=0.9,en-US;q=0.8\r\n
```

English – UK

- What is the IP address of your computer? Of the gaia.cs.umass.edu server?

Source	Destination
192.168.100.15	128.119.245.12
128.119.245.12	192.168.100.15

PC-> 192.168.100.15, Server-> 128.119.245.12

- What is the status code returned from the server to your browser?

```
Status Code: 200  
[Status Code Description: OK]
```

Code-> 200

- When was the HTML file that you are retrieving last modified at the server?

```
Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
```

- How many bytes of content are being returned to your browser?

```
Content-Length: 128\r\n
```

2. The HTTP Conditional GET/response interaction

- Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

```
Hypertext Transfer Protocol  
  GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n    Request Method: GET  
    Request URI: /wireshark-labs/HTTP-wireshark-file2.html  
    Request Version: HTTP/1.1  
    Host: gaia.cs.umass.edu\r\n    Connection: keep-alive\r\n    Upgrade-Insecure-Requests: 1\r\n    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/104.0.0.0 Safari/537.36\r\n    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8\r\n    Accept-Encoding: gzip, deflate\r\n    Accept-Language: en-GB,en-US;q=0.9,en;q=0.8\r\n    \r\n    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]
```

There is no If-Modified-Since header

- Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

Status Code: 200

[Status Code Description: OK]

Line-based text data: text/html (10 lines)

Yes, server has returned the content of the file

- Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT\r\n

Yes, second time it shows the If-Modified-Since header. It follows the value of last modified header from the 1st server response

- What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

Status Code: 304

[Status Code Description: Not Modified]

Response Phrase: Not Modified

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wire

The server respond 304-Not Modified, and the server did not return the content of the file, This tells the browser to use the cache and the body of the message is empty.

3. Retrieving Long Documents (LAB EX)

- How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill of Rights?

GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1\r\n

Request Method: GET

Request URI: /wireshark-labs/HTTP-wireshark-file3.html

Request Version: HTTP/1.1

1036 25.297131700 10.209.184.164 128.119.245.12 HTTP 552 GET /

1 GET Request and packet 1036 has GET message

- 2. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

HTTP/1.1 200 OK\r\n

Response Version: HTTP/1.1

Status Code: 200

[Status Code Description: OK

Response Phrase: OK

1201 25.688725100 128.119.245.12 10.209.184.164 HTTP 748 HTTP/1.1 200

Packet 1201 has a response status code and phrase

- 3. What is the status code and phrase in the response?

```

Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK

```

- 4. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

```

Transmission Control Protocol, Src Port: 80, Dst Port: 53076
  Source Port: 80
  Destination Port: 53076
  [Stream index: 38]
  [Stream Packet Number: 6]

```

6 packets containing TCP segments

4. HTML Documents with Embedded Objects (LAB EX)

- How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

853	59.741349900	192.168.100.15	128.119.245.12	HTTP	538 GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
864	60.140466200	192.168.100.15	128.119.245.12	HTTP	484 GET /pearson.png HTTP/1.1
911	60.964471000	192.168.100.15	2.56.99.24	HTTP	451 GET /8E_cover_small.jpg HTTP/1.1

3 HTTP GET request message (text, image and logo)

- Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

No.	Time	Source	Destination	Protocol	Length	Info
853	59.741349900	192.168.100.15	128.119.245.12	HTTP	538	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
856	60.004403900	128.119.245.12	192.168.100.15	HTTP	1362	HTTP/1.1 200 OK (text/html)
864	60.140466200	192.168.100.15	128.119.245.12	HTTP	484	GET /pearson.png HTTP/1.1
876	60.338432200	128.119.245.12	192.168.100.15	HTTP	640	HTTP/1.1 301 Moved Permanently (text/html)
911	60.964471000	192.168.100.15	2.56.99.24	HTTP	451	GET /8E_cover_small.jpg HTTP/1.1
1333	62.090461900	2.56.99.24	192.168.100.15	HTTP	288	HTTP/1.1 200 OK (JPEG JFIF image)

The downloading is serial as the server responds to the browser after every GET request.

5. HTTP Authentication (LAB EX)

- What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?

```

HTTP/1.1 401 Unauthorized\r\n
  Date: Sun, 08 Mar 2026 19:23:16 GMT\r\n
  Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.1 mod_fcgid/2.3.9 mod_perl/2
  WWW-Authenticate: Basic realm="wireshark-students only"\r\n

```

The server responded 401-> unauthorized access

- When your browser's sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?

```
Connection: keep-alive\r\n
Cache-Control: max-age=0\r\n
▶ Authorization: Basic c2FyYWZsaToxMjM0\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/
```

Authorization field was added

6. Capturing a Bulk TCP transfer from your computer to a Remote Server (LAB EX)

- What is the IP address and TCP port number used by the client computer (source) that is transferring the file to gaia.cs.umass.edu? To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the "details of the selected packet header window".

No.	Time	Source	Destination	Protocol	Length	Info
2809	167.415043900	192.168.100.15	128.119.245.12	HTTP	712	PO
2836	167.626280900	128.119.245.12	192.168.100.15	HTTP	838	HT


```
Frame 2809: Packet, 712 bytes on wire (5696 bits), 712 bytes captured
Ethernet II, Src: Intel_47:09:7d (7c:2a:31:47:09:7d), Dst: HuaweiTec
Internet Protocol Version 4, Src: 192.168.100.15, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 57545, Dst Port: 80, Seq: 1
Source Port: 57545
Destination Port: 80
[Stream index: 42]
```

IP: 192.168.100.15 and Port:57545

- What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection? If you have been able to create your own trace, answer the following question:

No.	Time	Source	Destination	Protocol	Length	Info
589	89.831638800	192.168.100.15	128.119.245.12	HTTP	535	GET /wiresharklabs/alice.txt HTTP/1.1
626	90.060351300	128.119.245.12	192.168.100.15	HTTP	665	HTTP/1.1 301 Moved Permanently (text/html)


```
Frame 589: Packet, 535 bytes on wire (4280 bits), 535 bytes captured (4280 bits) on interface \Device\NPF_{374D6330-DB64-4FC5-838E-D874FBA0C0}
Ethernet II, Src: Intel_47:09:7d (7c:2a:31:47:09:7d), Dst: HuaweiTechno_ab:44:8a (78:17:be:ab:44:8a)
Internet Protocol Version 4, Src: 192.168.100.15, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 57521, Dst Port: 80, Seq: 1, Ack: 1, Len: 481
Source Port: 57521
Destination Port: 80
[Stream index: 18]
[Stream Packet Number: 4]
```

Wireless LAN adapter WiFi:

```
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::bcaa:7d4c:6ca6:68b2%9
IPv4 Address. . . . . : 192.168.100.15
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.100.1
```

- What is the IP address and TCP port number used by your client computer (source) to transfer the file to gaia.cs.umass.edu?

```
Internet Protocol Version 4, Src: 192.168.100.15, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 57545, Dst Port: 80, Seq: 180390, Ack: 1, Len: 658
Source Port: 57545
Destination Port: 80
[Stream index: 42]
[Stream Packet Number: 164]
```